ON ORDOWSKI'S CHARACTERIZATION OF ODD PRIMES BY THE NUMBER OF SQUARES MODULO n: PRIME POWERS AND SEMIPRIMES, FORMALIZED IN LEAN 4

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Abstract. Let $a(n)$ be the number of squares modulo n (OEIS A000224). T. Ordowski conjectured that for $n > 1$, $n^2 \equiv 1 \pmod{a(n)(a(n)-1)}$ holds if and only if n is an odd prime. We prove the conjecture for every n that is even, a prime power, or a product of two distinct odd primes. For prime powers we use Stangl's formula for $a(p^k)$, which we also formalize, together with a congruence argument modulo p. For semiprimes we combine a size argument bounding the smaller prime by 61, a divisibility argument bounding the larger prime, and a finite check. All results are formalized in Lean 4 with Mathlib against the exact definition of a used in Google DeepMind's `formal-conjectures` repository. The development uses no `sorry`, no `native_decide`, and only the standard axioms.

1. Introduction

For $n \geq 1$ let $a(n) = \#\{k^2 \bmod n : 0 \leq k < n\}$ be the number of squares in $\mathbb{Z}/n\mathbb{Z}$ (OEIS A000224).

Conjecture 1 (Ordowski)**.** *For $n > 1$: n is an odd prime $\iff$ $n^2 \equiv 1 \pmod{a(n)(a(n)-1)}$.*

The conjecture appears as a comment on A000224 (T. Ordowski, Apr. 13, 2025), together with the remark that it holds up to $n = 10^8$ (M. Marcus). It is included as an open problem (`category research open`) in the `formal-conjectures` repository [1], file `FormalConjectures/OEIS/224.lean`. The direction "odd prime $\Rightarrow$ congruence" is immediate, since $a(p) = (p+1)/2$. The open direction is that no composite n satisfies the congruence.

Theorem 2. *Conjecture 1 holds for every $n > 1$ that is even, a prime power p^k ($k \geq 1$), or of the form $n = pq$ with $p \neq q$ odd primes.*

2. Preliminaries

Lemma 3. *$a(n) = \#\{x^2 : x \in \mathbb{Z}/n\}$; a is multiplicative; and $a(p) = (p+1)/2$ for odd primes p.*

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Proof. The first claim holds because reduction mod n is a bijection $\{0, \dots, n-1\} \to \mathbb{Z}/n$. If $\gcd(m, n) = 1$, the CRT ring isomorphism $\mathbb{Z}/mn \cong \mathbb{Z}/m \times \mathbb{Z}/n$ commutes with squaring, so it maps the set of squares onto the product of the two sets of squares. For an odd prime p, the squares are the values k^2 with $0 \le k \le (p-1)/2$, since $x^2 = (-x)^2$. These values are distinct: $k^2 = l^2$ forces $p \mid (k-l)(k+l)$, and $0 \le k + l < p$. □

3. The even case

If n is even, then $n^2 - 1$ is odd, while $a(n)(a(n) - 1)$ is a product of two consecutive integers and hence even. So $n^2 \not\equiv 1$. (If $a(n)(a(n) - 1) = 0$, the congruence would say $n^2 = 1$, which is impossible for $n > 1$.)

4. The semiprime case

Theorem 4. *Let $p < q$ be odd integers with $p \ge 3$ prime, and let $s = (p+1)(q+1)/4$. Then $s(s-1) \nmid (pq)^2 - 1$.*

Primality of q is not needed. By Lemma 3, $a(pq) = s$ for distinct odd primes, so Theorem 4 gives the semiprime case of Theorem 2.

Proof. Put $P = (p+1)(q+1)$, so that $16\, s(s-1) = P(P-4)$. Suppose $Q\, s(s-1) = (pq)^2 - 1$, so $Q\, P(P-4) = 16((pq)^2 - 1)$.

Step 1: $Q \le 15$. Expanding, $P(P-4) - ((pq)^2 - 1) = 2pq(p+q+1) - 4pq + (p+q+1)^2 - 4(p+q+1) + 1 > 0$. Hence $Q < 16$.

Step 2: $p \le 61$. By Step 1, $16((pq)^2 - 1) \le 15P(P-4) < 15P^2$. Suppose $p \ge 62$. Since $q > p$ we have $(q+1)p \le (p+1)q$, so $P^2 p^2 \le (p+1)^4 q^2$. Also $15(p+1)^4 \le 16p^4 - 16$ for $p \ge 62$. Together with $q \ge p$ this gives $15P^2 \le 16((pq)^2 - 1)$, a contradiction.

Step 3: $q + 1 \mid 16(p^2 - 1)$. Since $q + 1 \mid P$, we get $q + 1 \mid 16((pq)^2 - 1)$. Also $16((pq)^2 - 1) - 16(p^2 - 1) = 16p^2(q^2 - 1)$ is divisible by $q + 1$.

Step 4: finite check. For each prime $p \in [3, 61]$ and each divisor $d = q + 1$ of $16(p^2 - 1)$ with $q > p$ odd, direct computation shows $s(s-1) \nmid (pq)^2 - 1$. There are only finitely many such pairs. □

5. Odd prime powers

Lemma 5 (Stangl [3])**.** *For an odd prime p and $k \ge 2$, $a(p^k) = a(p^{k-2}) + \frac{1}{2}p^{k-1}(p-1)$. Consequently $2(p+1)\, a(p^k) = p^{k+1} + p + 2$ if k is even and $p^{k+1} + 2p + 1$ if k is odd.*

Proof. Let $N = p^k$. A square $x^2 \bmod N$ is divisible by p iff $p \mid x$. Writing $x = pt$ gives $(pt)^2 \bmod p^2 M = p^2(t^2 \bmod M)$ with $M = p^{k-2}$, so the squares divisible by p are exactly p^2 times the squares modulo M. For the squares prime to p: if $x, y \in [0, N/2)$ are prime to p and $N \mid (y - x)(y + x)$, then p cannot divide both factors (as p is odd and $p \nmid y$), so N divides one factor, forcing $x = y$. Replacing x by $N - x$, every unit square has such a representative. Hence the unit squares number $\varphi(N)/2$. The closed form follows by induction from $a(1) = 1$ and $a(p) = (p+1)/2$. □

Theorem 6. *If p is an odd prime, $k \geq 2$, and s satisfies the identity of Lemma 5, then $s(s-1) \nmid p^{2k}-1$. Consequently, $(p^k)^2 \not\equiv 1 \pmod{a(p^k)(a(p^k)-1)}$, and Conjecture 1 holds for all odd prime powers.*

Sketch. Let $X = p^k$ and $M = 2(p+1)$, and write $Q\,s(s-1) = X^2 - 1$. Multiplying by M^2 gives $Q(Ms)(Ms - M) = M^2(X^2 - 1)$. For even k, reducing modulo p gives $p \mid 4(p+1)^2$, which is impossible. For odd $k \geq 3$, a size estimate gives $1 \leq Q \leq 7$, and reducing modulo p gives $p \mid Q - 4$. If $Q = 4$, the equation becomes $(2p+1)X^2 = 2p^2X + p^2$, which is impossible since $X > p$. Otherwise $p = 3$ and $Q \in \{1, 7\}$, and both cases are ruled out by elementary inequalities. □

6. Lean formalization

The repository accompanying this paper (Lean `v4.35.0-rc4`, Mathlib tag `v4.35.0-rc4`) contains:

- `Defs.lean`: the definition of a, copied verbatim from `formal-conjectures`.
- `Count.lean`: `a_eq_aZ`, `aZ_mul` (CRT) and `aZ_prime`, i.e. Lemma 3.
- `Basic.lean`: `even_case` and `prime_direction`.
- `FiniteCheck.lean`: Step 4, checked by `decide +kernel` (kernel evaluation, no `native_decide`).
- `Semiprime.lean`: `pq_reduction` (Steps 1–3), `pq_case` and `semiprime_case`.
- `PrimePower.lean`: `card_Sq_split`, `two_mul_card_unit_Sq`, `stangl` (Lemma 5) and `prime_power_case`.
- `Main.lean`: `conjecture_main`, i.e. Theorem 2, stated in the exact form of `OeisA224.conjecture`.
- `PrimePowerCore.lean`: the arithmetic core of Theorem 6 (`pp_even`, `pp_odd`).

`#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound` for every theorem.

7. Novelty and related work

As of October 12, 2026, the OEIS entry lists the conjecture as open, and `formal-conjectures` marks it `research open`. Searches of the web, the repository's issues and pull requests, and AI-paper indexes found no proof, partial or complete. The ingredients (multiplicativity and Stangl's formula [3]; see also Finch–Sebah [4]) are classical. The even case is trivial. To our knowledge, the semiprime and prime-power cases are new, although the arguments are elementary. The squarefree case with three or more prime factors, and general odd composites, remain open. *Remark (not formalized).* The method of Step 3 still shows that if $n = mq$ satisfies the congruence, with q prime and $q \nmid m$, then $q + 1 \mid 4(m^2 - 1)$. For three or more prime factors, however, the size argument of Step 2 no longer bounds the second-smallest prime, so our method does not directly extend.

References

[1] Google DeepMind, *formal-conjectures*, `https://github.com/google-deepmind/formal-conjectures`, file `FormalConjectures/OEIS/224.lean`.
[2] OEIS Foundation, *A000224: Number of squares mod n*, `https://oeis.org/A000224`.
[3] W. D. Stangl, Counting squares in $\mathbb{Z}_n$, *Math. Mag.* **69** (1996), 285–289.
[4] S. R. Finch and P. Sebah, Squares and cubes modulo n, arXiv:math/0604465.
[5] The mathlib Community, The Lean mathematical library, CPP 2020.